Toolkit: Chemistry of the Earth

Practical Five: Chemical Reactions and Oxidation State

Objectives:

- Practice balancing chemical reactions relevant to minerals and rocks
- Practice calculations involving moles and molecular mass
- Investigate some important oxidation-reduction reactions in geochemical systems

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FEEDBACK

A. Balancing chemical reactions

For the following chemical reactions, determine whether or not the reaction is balanced. This means that for each element that appears in the equation, there must be the same number of atoms on both sides. If the reaction is not balanced, balance it!

1) Accumulations of sea shells on the ocean floor contain the mineral calcite (calcium carbonate). In a subduction zone, calcite is carried down the subducting slab. As it heats up, calcite reacts with quartz and breaks down to form wollastonite and carbon dioxide. The reaction is called decarbonation. The CO_2 gas released drives volcanic eruptions.

Calcite + Quartz = Wollastonite + Carbon dioxide

$CaCO_3$ + SiO_2 = $CaSiO_3$ + CO_2

Element	Calcite	Quartz	Total LHS	=	Wollastonite	Carbon dioxide	Total RHS
Ca	**1**		**1**		**1**		**1**
C	**1**		**1**			**1**	**1**
O	**3**	**2**	**5**		**3**	**2**	**5**
Si		**1**	**1**		**1**		**1**

The reaction is balanced as written.

2) In metamorphic rocks, muscovite reacts with quartz to form K-feldspar, an aluminosilicate mineral and water:

Muscovite + Quartz = K-feldspar + Sillimanite + Water

$KAl_2Si_3AlO_{10}(OH)_2$ + SiO_2 = $KAlSi_3O_8$ + Al_2SiO_5 + H_2O

Element	Muscovite	Quartz	Total LHS	=	K-feldspar	Sillimanite	Water	Total RHS
K	1		1		1			1
Al	3		3		1	2		3
Si	3	1	4		3	1		4
O	12	2	14		8	5	1	14
H	2		2				2	2

The reaction is balanced as written.

3) Another common reaction in metamorphic rocks involves the reaction of orthopyroxene and plagioclase to produce garnet, clinopyroxene and quartz. Orthopyroxene has very low calcium content (shown here as zero Ca), and clinopyroxene has equal amounts of calcium and magnesium in its formula.

$MgSiO_3$ + $CaAl_2Si_2O_8$ = $Mg_3Al_2Si_3O_{12}$ + $CaMgSi_2O_6$ + SiO_2

Orthopyroxene + Plagioclase = Garnet + Clinopyroxene + Quartz

Hint: Balance Mg first

Element	Orthopyroxene X4	Plagioclase	Total LHS	=	Garnet	Clinopyroxene	Quartz	Total RHS
Mg	1 / 4		1 / 4		3	1		4
Si	1 / 4	2	3 / 6		3	2	1	6
O	3 / 12	8	11 / 20		12	6	2	20
Ca		1	1			1		1
Al		2	2		2			2

The reaction is not balanced as written.

As written, both Ca and Al are balanced so we don't need to start on trying to balance these. Mg is the first element we should try to balance because it only appears in one mineral (Orthopyroxene) on the LHS. Multiplying Orthopyroxene by 4 gives the numbers in red. Just this single step balances the reaction:

4 $MgSiO_3$ + $CaAl_2Si_2O_8$ = $Mg_3Al_2Si_3O_{12}$ + $CaMgSi_2O_6$ + SiO_2

4) Another reaction in metamorphic rocks involves the reaction of talc and enstatite to produce anthophyllite:

$Mg_3Si_4O_{10}(OH)_2$ + $MgSiO_3$ = $Mg_7Si_8O_{22}(OH)_2$

Talc + Enstatite = Anthophyllite

Hint: Balance Mg first

Element	Talc	Enstatite **X4**	Total LHS	=	Anthophyllite	Total RHS
Mg	**3**	**1 / 4**	**4 / 7**		**7**	**7**
Si	**4**	**1 / 4**	**5 / 8**		**8**	**8**
O	**12**	**3 / 12**	**15 / 24**		**24**	**24**
H	**2**		**2**		**2**	**2**

The reaction is not balanced as written.

Starting with Mg, there are 4 atoms on the left and 7 on the right. To balance this, we could do one of two things: either (2 x Talc) + (1 x Enstatite) or (1 x Talc) + (4 x Enstatite). A quick glance at the other elements should show you that (1 x Talc) + (4 x Enstatite) is going to be better. One reason for this is that H is already balanced (2 each side) so multiplying Talc by 2 will unbalance the H.

Multiplying Enstatite by 4 results in the red numbers. Now the totals on the LHS equal those on the RHS so the reaction is now balanced:

$Mg_3Si_4O_{10}(OH)_2$ + 4 $MgSiO_3$ = $Mg_7Si_8O_{22}(OH)_2$

5) Yet another metamorphic reaction for those who like a challenge. It's OK to skip this.

$Ca_2Mg_5Si_8O_{22}(OH)_2$ + $CaCO_3$ + SiO_2 = $CaMgSi_2O_6$ + CO_2 + H_2O

Tremolite + Calcite + Quartz = Diopside + Carbon dioxide + Water

Element	Tremolite	Calcite **X3**	Quartz **X2**	Total LHS	=	Diospide **X5**	Carbon dioxide **X3**	Water	Total RHS
Ca	**2**	**1 / 3**		**3 / 5**		**1 / 5**			**1 / 5**
Mg	**5**			**5**		**1 / 5**			**1 / 5**
Si	**8**		**1 / 2**	**9 / 10**		**2 / 10**			**2 / 10**
O	**24**	**3 / 9**	**2 / 4**	**29 / 31 / 37**		**6 / 30**	**2 / 6**	**1**	**9 / 33 / 37**
H	**2**			**2**				**2**	**2**
C		**1 / 3**		**1 / 3**			**1 / 3**		**1 / 3**

Follow the following steps:

- C is balanced, so don't start with a C-bearing mineral (Calcite, CO_2).
- H is balanced, so don't start with tremolite or H_2O.
- Leave O until last.
- Mg only appears in one mineral on each side, so start with Mg. Multiply Diopside by 5. (Red)
- Next, we need to balance Ca and Si. We want to avoid changing Mg, H and C (which are all balanced), so avoid tremolite. Balance Si next. Multiply Quartz by 2. (Green)
- Now turn to Ca. Again, avoid tremolite. Multiply Calcite by 3. (Purple)

- Finally, we now see that C is no longer balanced. In order to balance C and O, multiply CO_2 by 3. (Orange)

The reaction is now balanced:

$$Ca_2Mg_5Si_8O_{22}(OH)_2 + 3\ CaCO_3 + 2\ SiO_2 = 5\ CaMgSi_2O_6 + 3\ CO_2 + H_2O$$

B. Moles and Molecular Mass

The Avogadro constant is the number of constituent particles, usually atoms or molecules, that are contained in one mole of the substance. (Number of molecules per mole.)

Avogadro constant: 6.022×10^{23}

Example: 12 g of ^{12}C has 6.022×10^{23} carbon atoms

1) Calculate the molecular mass of pyrite, FeS_2, given that the atomic mass of Fe is 56, and the atomic mass of S is 32.

Molecular mass of FeS_2 = 56 + (2*32) = 120

What is the mass of one mole of pyrite? 120 g

What is the volume of one mole of pyrite? (Density of pyrite = 5 $g.cm^{-3}$)

Density = Mass / Volume 5 = 120 / Volume, Volume = 24 cm^3

2) Return to the equation in question A1 above:

Calcite	+	Quartz	=	Wollastonite	+	Carbon dioxide
$CaCO_3$	+	SiO_2	=	$CaSiO_3$	+	CO_2

Calculate the molecular mass of each mineral in this equation.
Atomic mass numbers: C = 12, O = 16, Si = 28, Ca = 40.

Molecular mass of calcite: 40 + 12 + (3*16) = 100
Molecular mass of quartz: 28 + (2*16) = 60
Molecular mass of wollastonite: 40 + 28 + (3*16) = 116
Molecular mass of CO_2: 12 + (2*16) = 44

What do you notice about the total molecular mass on each side of the equation?
The molecular masses balance on each side of the equation: 100 + 60 = 160 and 116 + 44 = 160.

What is the mass of 1 mole of calcite? 100 g

If we start with a mass of 1 kg of calcite, what mass of CO_2 will be produced through this reaction?

1 kg calcite = 10 moles
1 mole of calcite produces one mole of CO_2, so 10 moles of calcite produces 10 moles of CO_2
Mass of 10 moles of CO_2 = 10 x 44 = 440 g

How many atoms of carbon are present in the CO_2 that is produced?

10 moles of CO_2 contain 10 moles of carbon. The total number of atoms of carbon is ten times Avogadro's constant, i.e. 10 * (6.022×10^{23}) = 6.022×10^{24} atoms.

C. Oxidation – Reduction Reactions

In this section we will look at some important oxidation-reduction (redox) reactions in geochemical systems.

a) Formation of uranium minerals (see McSween, p. 132, Worked Problem 7.5)

Much of the world's mineable uranium occurs as uraninite (UO_2) in sandstone-hosted deposits. To explain this, we will look at some important redox reactions.

The primary source of uranium is granitic rocks, that can contain up to 15 ppm uranium. When a granite is weathered by reaction with water, uranium dissolves in the water as U^{6+} ions, and then it is readily carried by the water in the form of complex ions. In acidic water, the uranyl ion is stable:

UO_2^{2+} $\quad$ $1*U^{6+}$ and $2*O^{2-}$ = 6+ + 4- = 2+

In alkaline water, U^{6+} may be transported as the complex ion:

$UO_2(CO_3)_3^{4-}$ $\quad$ $1*U^{6+}$ and $2*O^{2-}$ and $3*(CO_3)^{2-}$ = 6+ + 4- + 6- = 4-

Show how the charge on the two ions above are a balance of U^{6+} ions, O^{2-} ions, and CO_3^{2-} ions.

If the water also begins to dissolve iron as Fe^{2+} ions, the following reaction can occur:

$$UO_2(CO_3)_3^{4-} + 2Fe^{2+} + 3H_2O = UO_2 + 2Fe(OH)_3 + 3CO_2$$

- What is the valence of U in UO_2? U^{4+}. One U^{4+} and two O^{2-} = overall charge of zero.
- Is U^{6+} oxidized or reduced in this reaction? (explain)

  U^{6+} is reduced to U^{4+} (gain of electrons)
- Is Fe^{2+} oxidized or reduced in this reaction? (explain)

  Fe^{2+} is oxidized to Fe^{3+} (loss of an electron)

The reaction takes place within a sandstone. Both UO_2 (uraninite) and $Fe(OH)_3$ (iron hydroxide or ferric hydroxide) are insoluble in water, so both of these minerals precipitate in the sandstone.

Another important U ore mineral is coffinite, $USiO_4$. What is the valence of U in this mineral?

U^{4+} (charge balanced with $[SiO_4]^{4-}$ – tetrahedral silica ion)

Guess (before you google) how this mineral got its name:

Coffinite is named after American geologist Reuben Clare Coffin (1886–1972)

b) Oxidation of pyrite

An important redox reaction in nature is the conversion of pyrite (FeS_2) into a precipitate of $Fe(OH)_3$ and SO_4^{2-} ions in solution. This occurs when pyrite is exposed to atmospheric oxygen. The basic reaction can be written as:

$$FeS_2 + O_2 = Fe(OH)_3 + SO_4^{2-}$$

In pyrite, the oxidation state of iron is +2. Write a balanced half-reaction showing the change in oxidation state of iron in this equation, and the electrons produced:

$Fe^{2+} = Fe^{3+} + e^-$

Is this an oxidation or a reduction reaction? Oxidation (loss of electrons)

Write a balanced half-reaction showing the change in oxidation state of sulphur in this equation, and the electrons produced:

$2\ S^- = 2\ S^{6+} + 14\ e^-$

Is this an oxidation or a reduction reaction? Oxidation (loss of electrons)

Overall, in this reaction how many moles of electrons are produced per mole of pyrite that reacts?

One electron per Fe atom, 7 per S atom. Two S atoms give 14 electrons. Total = 15 moles of electrons per mole of pyrite.

In order to balance the reaction, we need to add water on the left side of the reaction and hydrogen ions on the right side:

$4\ FeS_2 + 15\ O_2 + 14\ H_2O = \square\ Fe(OH)_3 + \square\ SO_4^{2-} + \square\ H^+$

$4\ FeS_2 + 15\ O_2 + 14\ H_2O = 4\ Fe(OH)_3 + 8\ SO_4^{2-} + 16\ H^+$

Balance this reaction by filling in the boxes on the right hand side. Make sure the charges balance as well as the elements.

Pyrite is ultimately converted to the mineral limonite:

$2\ Fe(OH)_3 = Fe_2O_3 \bullet 3H_2O$

Acidic solutions have high concentrations of hydrogen ions. You might recognize that H_2SO_4 is sulphuric acid. Mining operations that expose formerly buried pyrite results in acidification of groundwater, producing acid mine waters that can be harmful to local ecosystems.

Decomposition of pyrite is accelerated by the activity of bacteria such as *Thiobacillus ferroxidans* that oxidize Fe^{2+} to Fe^{3+}, for example by enzymatic oxidation. The pyrite decomposition reaction then takes place in the absence of oxygen:

$FeS_2 + 14\ Fe^{3+} + 8\ H_2O = 15\ Fe^{2+} + 2\ SO_4^{2-} + 16\ H^+$

Balance this reaction by filling in the boxes on the right hand side. Make sure the charges balance as well as the elements.

c) Banded Iron Formations (BIFs) (see McSween p. 164)

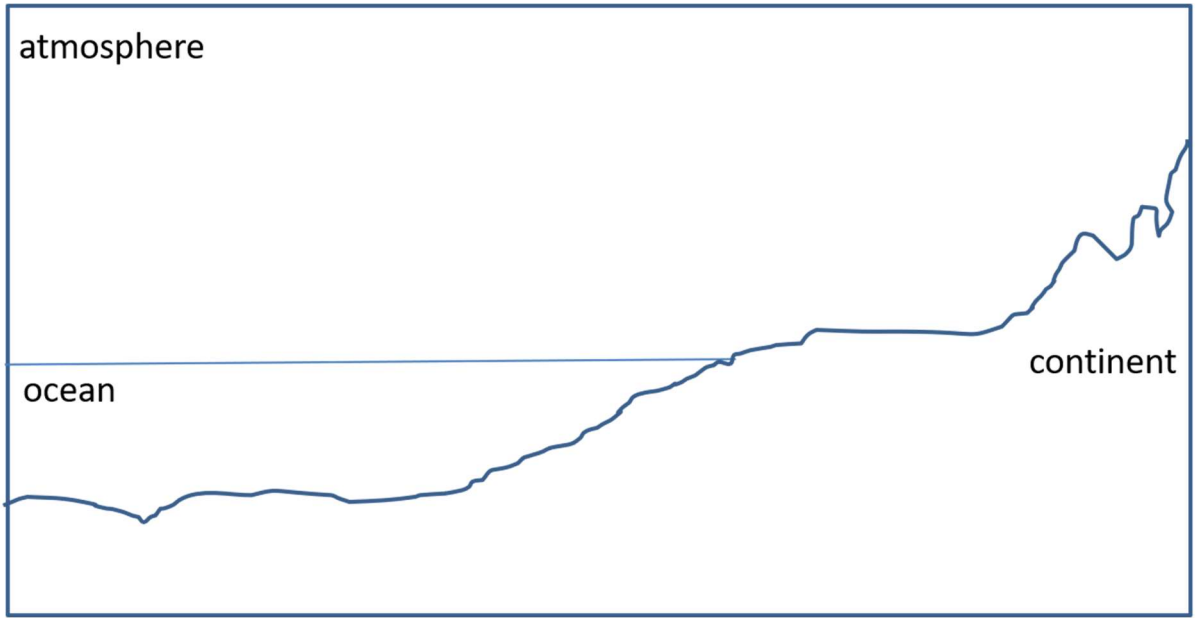
In the early Earth, there was little free oxygen in the atmosphere. Photosynthetically generated O_2, produced by cyanobacteria, first appeared around 2.7 billion years ago. For the next 400 million years, oxygen levels rose only slowly. Part of the evidence for this is the presence of uraninite (see (a) above) and pyrite (see (b) above) in sedimentary rocks from that *time*. Both of these minerals indicate reducing conditions. Cyanobacteria were producing O_2, but that O_2 was consumed

efficiently in the production of banded iron formations. Extensive BIFs were formed during this period, but they have not been formed since.

Oxygen released by cyanobacteria in surface ocean waters in the late Precambrian oxidized Fe^{2+} that was dissolved in the ocean, producing Fe^{3+} ions which form insoluble oxides (Fe_2O_3), carbonates and silicates. These minerals formed deposits on the sea floor. Fe^{2+} in the oceans was most likely derived from seafloor volcanism. As the rate of volcanism slowed down, the Fe^{2+} concentration in the oceans declined, and O_2 levels in the atmosphere began to rise. BIFs represent the transition to an oxygen-rich atmosphere that ended around 2.3 billion years ago.

Banded iron formations, or BIFs, are a major resource as the main ore for mining iron – 90% of the world's commercial iron supply.

On the outline sketch below, illustrate the processes described in the text above. An aspect of BIFs that is not well understood is why the iron / chert banding is cyclical (chert is SiO_2). Can you propose a hypothesis to explain this observation?



Your sketch should show the items highlighted in yellow:

- Cyanobacteria in the ocean, at the surface
- Uraninite and pyrite in sedimentary rocks on the land (at the base of the mountains)
- Oxygen from cyanobacteria combining with Fe^{2+} in the ocean to form Fe_2O_3 which sinks to the ocean floor and forms a BIF deposit
- A submarine volcano, releasing Fe^{2+} ions into the ocean
- Oxygen released from the ocean into the atmosphere

Also on the sketch, write the time that your sketch refers to.